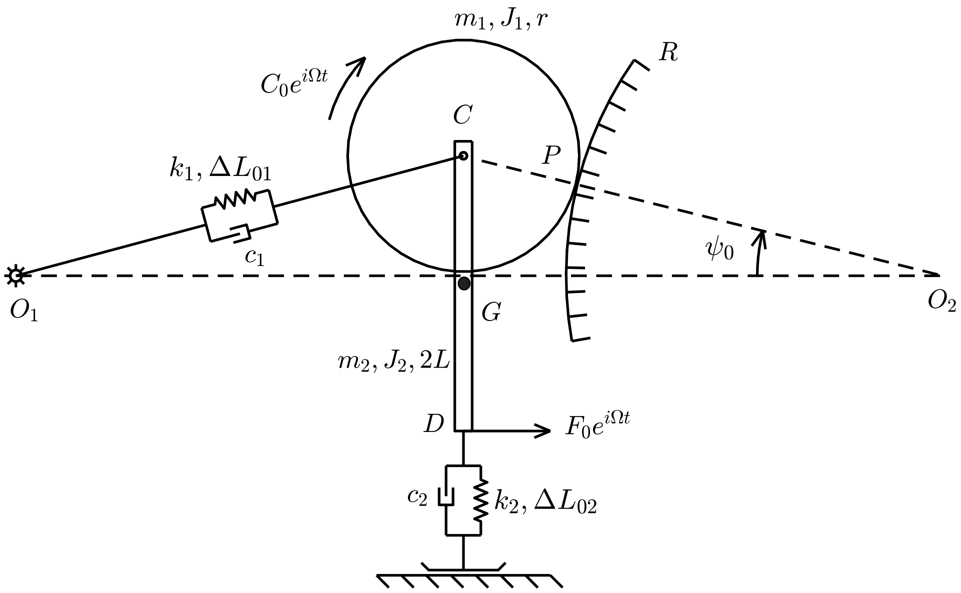
Dynamics of Mechanical Systems

June 23rd, 2020 – standard problem **(grading up to 30/30)**



m_1 = 6 kg	k_1 = 5000 N/m
m_2 = 4 kg	k_2 = 5000 N/m
J_1 = 0.12 kgm^2	ΔL_{01} = -0.0216 m
J_2 = 0.1 kgm^2	ΔL_{02} = 0.002 m
L = 0.25 m	c_1 = 2 Ns/m
r = 0.2 m	c_2 = 2 Ns/m
R = 1.0 m	$\psi_0 = \pi/6$

The mechanical system above lies in the vertical plane and is shown in its equilibrium position. The disk rolls without sliding on the circular guide. In the static configuration $O_1C = O_2C$.

The students are requested to:

0. Write on top of each page their surname, name and ID number
1. Write the linearized equations of motion around the given equilibrium configuration; in response to this point, upload on MS-Forms one single file in pdf format containing:
   - i. the drawing of the system marked with all kinematic variables used to solve the exam;
   - ii. The vector of independent coordinates chosen to write the system's equations of motion;
   - iii. The Jacobian matrices [Λm], [Λc], [Λk], with evidence of the procedure used to derive these results;
   - iv. The gravitational & second order stiffness, with evidence of the procedure used to derive these results;
2. Compute the natural frequencies and modes of vibrations;
3. Compute the Frequency Response Function (in the range 0 – 5 Hz, step = 0.01 Hz) applying the harmonic torque $C_0 e^{i\Omega t}$. Output: the horizontal component of the displacement of point C and the rotation of the bar. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your matriculation number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
4. Compute the Frequency Response Function (in the range 0 – 5 Hz, step = 0.01 Hz) applying the harmonic force $F_0 e^{i\Omega t}$. Output: the elastic and damping force transmitted by the spring/damper system (k_1, c_1) and the horizontal component of the displacement of point G. Save the graphical results as SNxxx3.fig and SNxxx4.fig
5. Compute the Frequency Response Function (in the range 0 – 5 Hz, step = 0.01 Hz) applying the harmonic force $F_0 e^{i\Omega t}$. Output: the normal (N) and tangential (T) dynamic components of the contact force exchanged between the disk and the ugide in point P. Save the graphical result as SNxxx5.fig SNxxx6.fig. In response to this point, also include in the pdf file uploaded on MS-Forms:
   - v. A drawing and the symbolic equations required to derive the normal (N) and tangential (T) components of the contact force.

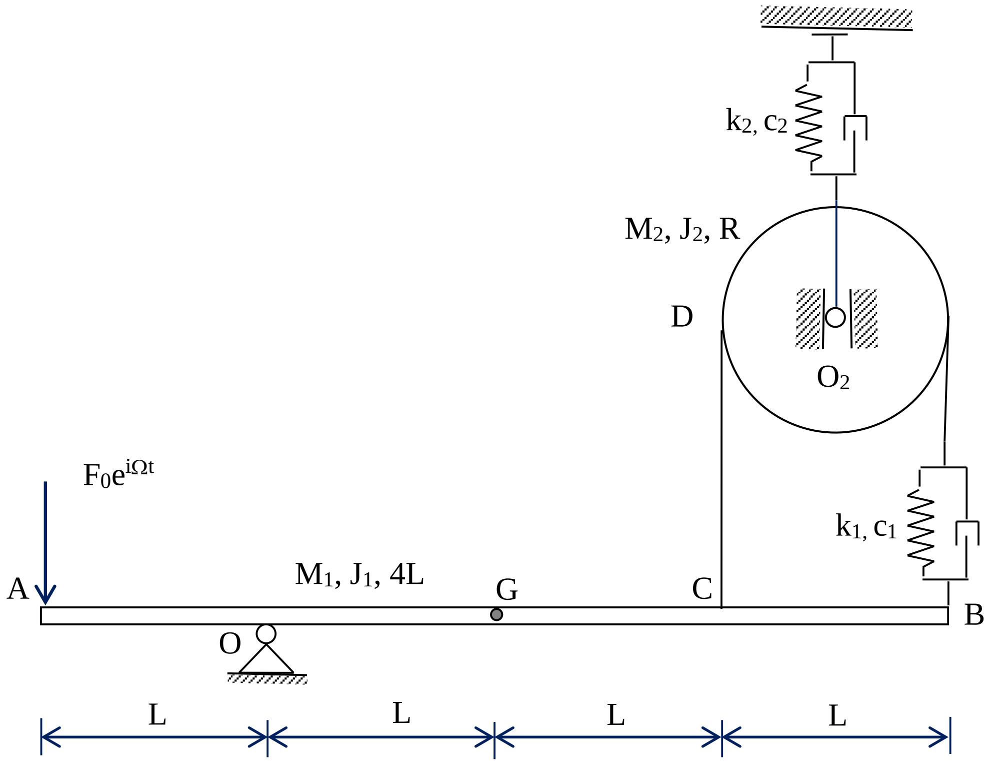
<u>The answer to points 2 – 5 shall be provided by delivering the MATLAB script and .fig files as a compressed folder uploaded on BeeP , see instructions below.</u>

<u>Note</u>: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

1. Connect to the site https://beep.metid.polimi.it and enter the portal using your credentials;
3. Select section **<u>Homework</u>** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder ***Test June 23rd 2020 - standard***
5. Click on button Add and select Single document;
6. Browse the compressed file SNxxx.zip using the button Browse… (or Sfoglia…)
7. Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.

Dynamics of Mechanical Systems

June 23rd, 2020 – simplified problem (**grading up to 23/30**)



$M_1 =$ 4 kg	$J_1 = 0.1$ kgm^2	$L =$ 0.25 m	$k_1 =$ 1000 N/m	$c_1 = 1$ Ns/m
$M_2 =$ 1 kg	$J_2 = 0.1$ kgm^2	$R =$ L/2	$k_2 =$ 1000 N/m	$c_2 = 1$ Ns/m

The mechanical system in the picture lies in the vertical plane and is shown in its equilibrium position. The students are requested to:

0. Write on top of each page their surname, name and ID number
1. Write the linearized equations of motion around the given static configuration; in response to this point, upload on MS-Forms one single file in pdf format containing:
   i. the drawing of the system marked with all kinematic variables used to solve the exam;
   ii. The vector of independent coordinates chosen to write the system's equations of motion;
   iii. The Jacobian matrices [Λm], [Λc], [Λk], with evidence of the procedure used to derive these results;
2. Compute the natural frequencies and modes of vibrations;
3. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic force $F_0 e^{i\Omega t}$. Output: the bar rotation and the disk rotation. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your identification number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic force $F_0 e^{i\Omega t}$. Output: the tension in the CD string. Save the graphical results as SNxxx3.fig. In response to this point, also include in the pdf file uploaded on MS-Forms:
   iv. A drawing and the symbolic equation required to derive the tension in the CD string

The answer to points 2 – 4 shall be provided by delivering the MATLAB script and .fig files as a compressed folder uploaded on BeeP , see instructions below.

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

1. Connect to the site https://beep.metid.polimi.it and enter the portal using your credentials;
3. Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder ***Test June 23rd 2020 - simplified***
5. Click on button Add and select Single document;
6. Browse the compressed file SNxxx.zip using the button Browse… (or Sfoglia…)
7. Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.